Math 124 I - Winter 2007
Mid-Term Exam Number Two
February 20, 2007
Answers

Version A (problem 1(a) starts $y = 3x^6 ...$)

1. (a) $\dfrac{dy}{dx} = 18x^5 + \dfrac{\sqrt{5}}{2}x^{-1/2} + 36x^{-4} - 4\sin 4x$

   (b) $\dfrac{dy}{dx} = (15x^4+1)e^{3x^5+x}\sin(2x^2+e^x) + e^{3x^5+x}\cos(2x^2+e^x)(4x+e^x)$

   (c) $\dfrac{(e^x(x^5-x-1)+e^x(5x^4-1))(x^2+3) - 2xe^x(x^5-x-1)}{(x^2+3)^2}$

2. (a) $\dfrac{dy}{dx} = \dfrac{2-y+\sec^2(x+y)}{x-3-\sec^2(x+y)}$

   (b) $\dfrac{dy}{dx} = \dfrac{1}{x+2} + \dfrac{1}{x+1} - \dfrac{1}{x-1} - \dfrac{1}{x-3}$

   (c) $x^{(x+1)(x+2)}\left(\ln x(2x+3) + \dfrac{1}{x}(x+1)(x+2)\right)$

3. (a) $\dfrac{dr}{dt} = 0.9333$ cm/hr (b) -2

4. The point is $(-\frac{27}{8}, \frac{9}{4})$.

5. $y = -x - 2$ and $y = -\frac{25}{9}x + \frac{10}{3}$.

Version B (problem 1(a) starts $y = 5x^7 ...$)

1. (a) $\dfrac{dy}{dx} = 35x^6 + \dfrac{\sqrt{3}}{2}x^{-1/2} + 32x^{-5} - 2\sin 2x$

   (b) $\dfrac{dy}{dx} = e^{7x^2+x}(14x+1)\sin(4x^3+e^x) + e^{7x^2+x}\cos(4x^3+e^x)(12x^2+e^x)$

   (c) $\dfrac{dy}{dx} = \dfrac{(e^x(x^4-x-3)+e^x(4x^3-1))(5x^2+1) - 10xe^x(x^4-x-3)}{(5x^2+1)^2}$

2. (a) $\dfrac{dy}{dx} = \dfrac{7-y+\sec^2(x+y)}{x-5-\sec^2(x+y)}$

   (b) $\dfrac{dy}{dx} = \dfrac{1}{x+2} + \dfrac{1}{x+1} - \dfrac{1}{x-1} - \dfrac{1}{x-3}$

   (c) $\dfrac{dy}{dx} = x^{(x+1)(x+2)}\left((2x+4)\ln x + \dfrac{1}{x}(x+1)(x+3)\right)$

3. (a) -1.0 cm/hr (b) -2

4. The point is $(-\frac{27}{8}, \frac{9}{4})$.

5. $y = -x - 2$ and $y = -\frac{25}{9}x + \frac{10}{3}$.